The Singular Value Decomposition

A Rigorous Development from First Principles

culminating in the Eckart–Young–Mirsky Theorem

A self-contained graduate treatment

every theorem motivated, every step proved

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Preface

This book develops the Singular Value Decomposition (SVD) and its central optimality property — the *Eckart–Young–Mirsky best rank-k approximation theorem* — in the rigorous, build-everything-from-the-ground-up style of a graduate linear-algebra text. The guiding principle is simple and uncompromising:

Nothing is asserted that has not first been proved.

We deliberately avoid the phrases "it is easy to see," "clearly," and "it can be shown." When a theorem depends on another result, that result is stated and proved *first*. The reader is assumed to know only the mechanics of matrix multiplication, transpose, inverse, and determinants. Every other object — span, basis, rank, inner product, orthogonal projection, symmetric matrices, the Spectral Theorem, positive semidefiniteness, matrix norms — is introduced from scratch and proved before it is used.

Each chapter closes with a **summary** and a set of **exercises**. Throughout, shaded boxes separate the formal mathematics from *motivation*, *geometric interpretation*, *machine-learning interpretation*, and *common misconceptions*, so that intuition supports the proofs without ever substituting for them.

The logical dependency of the chapters is strictly linear:

$$\begin{gathered}\text{rank} \;\to\; \text{orthogonality} \;\to\; \text{symmetric matrices} \;\to\; \text{Spectral Theorem} \\ \to\; \text{PSD} \;\to\; \text{SVD} \;\to\; \text{norms} \;\to\; \text{Eckart–Young–Mirsky.}\end{gathered}$$

Read it in order; each chapter is a prerequisite for the next.

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Chapter 1

Motivation: Why the SVD Exists

Motivation & intuition

Eigenvalue decomposition is the most powerful tool in elementary linear algebra: it turns the action of a matrix into independent scalar stretches along special directions. But it has two fatal restrictions — it needs the matrix to be *square*, and even then it may fail to exist (the matrix may not be diagonalizable). Data matrices are almost never square, and we cannot afford a tool that sometimes does not exist. The SVD is the repair: a decomposition that exists for *every* matrix, square or rectangular, and that recovers all the computational power of the eigenbasis.

1.1 The eigenvalue decomposition and its limits

For a square matrix $A \in \mathbb{R}^{n\times n}$ the eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$ identifies directions $\mathbf{v}$ along which A acts as pure scaling. When A possesses n linearly independent eigenvectors, stacking them as the columns of V gives the *eigenvalue decomposition* (EVD)

$$A = V\Lambda V^{-1}, \qquad \Lambda = \operatorname{diag}(\lambda_1,\dots,\lambda_n),$$

which converts repeated multiplication by A into repeated scalar multiplication. This is the engine behind iterative methods, differential equations, and Markov chains.

The decomposition demands two things, and *both* can fail:

(i) **Squareness.** The equation $A\mathbf{v} = \lambda\mathbf{v}$ requires $A\mathbf{v}$ and $\mathbf{v}$ to lie in the same space. If $A \in \mathbb{R}^{m\times n}$ with $m \neq n$, then $\mathbf{v} \in \mathbb{R}^n$ but $A\mathbf{v} \in \mathbb{R}^m$, and the equation is not even well posed. Rectangular matrices simply *cannot* have eigenvectors.

(ii) **Diagonalizability.** Even square matrices may lack a full set of independent eigenvectors. The matrix $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ has the single eigenvalue 1 but only a one-dimensional eigenspace, so no V^{-1} exists.

1.2 What the SVD promises

The SVD circumvents both problems by abandoning the demand for a *single* basis in which A is diagonal. Instead it allows *two* orthonormal bases — one in the input space $\mathbb{R}^n$, one in the output space $\mathbb{R}^m$ — linked so that A maps one to the other with pure scaling in between. We will prove (Chapter 7) that *every* real matrix $A \in \mathbb{R}^{m\times n}$ factors as

$$A = U\Sigma V^{\top},$$

with $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ orthogonal and Σ a nonnegative diagonal matrix. No squareness, no diagonalizability hypothesis — the factorization always exists.

1.3 Why anyone outside pure mathematics should care

The SVD is arguably the single most useful matrix factorization in applied science. A non-exhaustive list of where its optimality (the subject of Chapter 9) is doing the work:

Principal Component Analysis (PCA).
The directions of maximal variance in a data cloud are exactly the right singular vectors of the centred data matrix (Chapter 10). Dimensionality reduction *is* truncated SVD.

Image compression.
A grayscale image is a matrix of pixel intensities. Because its singular values decay rapidly, a few dozen rank-one layers reproduce the picture while storing a small fraction of the numbers.

Latent Semantic Analysis (LSA).
A term–document matrix, truncated by SVD, exposes latent "topics" as singular directions, mapping synonyms to nearby vectors.

Recommendation systems.
A user–item rating matrix is modelled as approximately low rank; the missing entries are predicted by its best low-rank approximation.

LoRA (Low-Rank Adaptation).
Fine-tuning updates to enormous neural networks are empirically near low rank, so they are stored in factored form $\Delta W = BA$ with B, A thin — a direct application of the low-rank idea, slashing trainable parameters by orders of magnitude.

Numerical stability.
The SVD gives the most reliable computation of rank, pseudoinverse, and least-squares solutions, precisely because it never forms the condition-number-squaring product $A^{\mathsf{T}}A$.

Each of these rests on the theorem at the heart of this book: among all matrices of rank at most k, the truncated SVD is provably the closest to A. Everything we build in Chapters 2–8 is in service of proving that statement cleanly in Chapter 9.

Summary

EVD is powerful but restricted to square, diagonalizable matrices. The SVD removes both restrictions by using paired orthonormal bases in the input and output spaces. It exists universally and underlies PCA, compression, LSA, recommender systems, and LoRA. The remainder of the book proves the SVD exists and that its truncation is optimal.

Exercise 1.1. Exhibit a 2×2 real matrix that has no real eigenvalues, and verify directly that it nonetheless has a real SVD (you may use a computer for the arithmetic; we prove existence in general later).

Exercise 1.2. Explain in one paragraph why the equation $A\mathbf{v} = \lambda\mathbf{v}$ is dimensionally inconsistent when $A \in \mathbb{R}^{3\times 5}$.

Exercise 1.3. The matrix $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ is square. Show it has only a one-dimensional eigenspace, hence no EVD of the form $V\Lambda V^{-1}$ with diagonal Λ exists.

Chapter 2

Linear-Algebra Prerequisites: Rank

We develop only what the SVD requires, but we develop it completely. Throughout, vectors are columns and $\mathbb{R}^n$ carries the usual operations.

2.1 Linear independence, span, basis, dimension

Definition 2.1 (Linear combination, span)**.** A *linear combination* of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ is any vector $\sum_{i=1}^k c_i \mathbf{v}_i$ with $c_i \in \mathbb{R}$. Their *span* is the set of all such combinations, $\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{\sum_i c_i \mathbf{v}_i : c_i \in \mathbb{R}\}$, which is a subspace of $\mathbb{R}^n$.

Definition 2.2 (Linear independence)**.** The vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are *linearly independent* if the only scalars solving $\sum_{i=1}^k c_i \mathbf{v}_i = \mathbf{0}$ are $c_1 = \cdots = c_k = 0$. Otherwise they are *linearly dependent*.

Definition 2.3 (Basis, dimension)**.** A *basis* of a subspace $W \subseteq \mathbb{R}^n$ is a linearly independent set that spans W. The number of vectors in a basis is the *dimension* $\dim W$; we take as known from elementary linear algebra that every basis of a given subspace has the same size, so $\dim W$ is well defined.

Lemma 2.4 (Coordinates are unique in a basis)**.** *If $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis of W then every $\mathbf{w} \in W$ has a* unique *representation $\mathbf{w} = \sum_i c_i \mathbf{v}_i$.*

Proof. Existence holds because the basis spans W. For uniqueness, suppose $\sum_i c_i \mathbf{v}_i = \sum_i d_i \mathbf{v}_i$. Subtracting gives $\sum_i (c_i - d_i)\mathbf{v}_i = \mathbf{0}$, and linear independence forces $c_i - d_i = 0$ for all i, i.e. $c_i = d_i$. ■

2.2 The four fundamental subspaces and rank

Definition 2.5 (Column space, row space, null space)**.** For $A \in \mathbb{R}^{m \times n}$:

- the *column space* $\operatorname{col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$ is the span of the columns of A;
- the *row space* $\operatorname{row}(A) = \operatorname{col}(A^\mathsf{T}) \subseteq \mathbb{R}^n$ is the span of the rows;
- the *null space* $\operatorname{null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$.

Definition 2.6 (Rank: three candidate definitions)**.** For $A \in \mathbb{R}^{m \times n}$ define

$$r_{\text{col}} = \dim \operatorname{col}(A), \qquad r_{\text{row}} = \dim \operatorname{row}(A), \qquad r_{\text{fac}} = \min\{k : A = YZ^\mathsf{T},\ Y \in \mathbb{R}^{m \times k},\ Z \in \mathbb{R}^{n \times k}\}.$$

These three numbers coincide. We prove it, because the equality ("row rank = column rank") is used repeatedly later and should not be taken on faith.

Theorem 2.7 (Equivalence of the rank definitions)**.** *For every $A \in \mathbb{R}^{m\times n}$, $r_{\text{col}} = r_{\text{row}} = r_{\text{fac}}$. The common value is the* rank $\operatorname{rank}(A)$.

Proof. **Step 1: $r_{\text{col}} = r_{\text{fac}}$.** Let $r = r_{\text{col}}$ and pick a basis $\mathbf{b}_1,\dots,\mathbf{b}_r$ of $\operatorname{col}(A)$; assemble it as the columns of $Y = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_r] \in \mathbb{R}^{m\times r}$. Each column $\mathbf{a}_j$ of A lies in $\operatorname{col}(A)$, so by Lemma 2.4 it has coordinates $\mathbf{a}_j = \sum_{i=1}^r z_{ji}\mathbf{b}_i = Y\mathbf{z}_j$ where $\mathbf{z}_j = (z_{j1},\dots,z_{jr})^\top$. Stacking the $\mathbf{z}_j$ as rows of a matrix $Z \in \mathbb{R}^{n\times r}$ gives $A = YZ^\top$. Hence a factorization of inner dimension r exists, so $r_{\text{fac}} \le r$. Conversely, if $A = YZ^\top$ with $Y \in \mathbb{R}^{m\times k}$ then every column of A is $A\mathbf{e}_j = Y(Z^\top\mathbf{e}_j) \in \operatorname{col}(Y)$, so $\operatorname{col}(A) \subseteq \operatorname{col}(Y)$ and $r_{\text{col}} \le \dim\operatorname{col}(Y) \le k$. Taking $k = r_{\text{fac}}$ yields $r_{\text{col}} \le r_{\text{fac}}$. Combining, $r_{\text{col}} = r_{\text{fac}}$.

Step 2: $r_{\text{row}} = r_{\text{fac}}$. Apply Step 1 to $A^\top$: its column space is $\operatorname{row}(A)$, and $A^\top = ZY^\top$ is a factorization of inner dimension r_{fac} (the same Y, Z as above, transposed). Therefore $r_{\text{row}} = \dim\operatorname{col}(A^\top)$ equals the minimal inner dimension of a factorization of $A^\top$, which is r_{fac} because transposing a factorization of A gives one of $A^\top$ of the same inner dimension and vice versa.

Hence all three numbers agree. ■

Remark 2.8. Theorem 2.7 is the statement, familiar from a first course but rarely proved there, that *the maximal number of independent columns equals the maximal number of independent rows*. We will use the factorization form $A = YZ^\top$ (Figure "long-skinny times short-long") constantly.

2.3 Rank-0, rank-1, rank-2, rank-k

Definition 2.9 (Rank-k matrix)**.** *A has rank k if $\operatorname{rank}(A) = k$.* Equivalently (Theorem 2.7) A is a sum of k rank-one matrices but not of $k-1$.

The only rank-0 matrix is the zero matrix. Rank-one matrices have an exceptionally clean description, which is the workhorse of the entire theory.

Theorem 2.10 (Characterization of rank-one matrices)**.** *A matrix $A \in \mathbb{R}^{m\times n}$ has rank 1 if and only if $A = \mathbf{u}\mathbf{v}^\top$ for some nonzero vectors $\mathbf{u} \in \mathbb{R}^m$, $\mathbf{v} \in \mathbb{R}^n$.*

Proof. ($\Leftarrow$) Suppose $A = \mathbf{u}\mathbf{v}^\top$ with $\mathbf{u},\mathbf{v} \neq \mathbf{0}$. Column j of A is $A\mathbf{e}_j = \mathbf{u}(\mathbf{v}^\top\mathbf{e}_j) = v_j\mathbf{u}$, a scalar multiple of the fixed nonzero vector $\mathbf{u}$. Thus every column lies in $\operatorname{span}\{\mathbf{u}\}$, so $\operatorname{col}(A) \subseteq \operatorname{span}\{\mathbf{u}\}$, giving $\dim\operatorname{col}(A) \le 1$. Since some $v_j \neq 0$, the column $v_j\mathbf{u} \neq \mathbf{0}$, so $A \neq 0$ and $\dim\operatorname{col}(A) \ge 1$. Hence $\operatorname{rank}(A) = 1$.

($\Rightarrow$) Suppose $\operatorname{rank}(A) = 1$. Then $\operatorname{col}(A)$ is one-dimensional; let $\mathbf{u}$ be a nonzero spanning vector, so $\operatorname{col}(A) = \operatorname{span}\{\mathbf{u}\}$. Each column $\mathbf{a}_j \in \operatorname{col}(A)$ is a multiple $\mathbf{a}_j = v_j\mathbf{u}$ for a unique scalar v_j (Lemma 2.4). Setting $\mathbf{v} = (v_1,\dots,v_n)^\top$ gives, column by column, $A = [v_1\mathbf{u} \ \cdots \ v_n\mathbf{u}] = \mathbf{u}\mathbf{v}^\top$. As $A \neq 0$, $\mathbf{v} \neq \mathbf{0}$. ■

Example 2.11. $A = \left(\begin{smallmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{smallmatrix}\right)$ has every row a multiple of $(2,4,6)$ and every column a multiple of $(2,1)^\top$. Indeed $A = \left(\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 2 & 3 \end{smallmatrix}\right) = \mathbf{u}\mathbf{v}^\top$, confirming $\operatorname{rank}(A) = 1$.

Remark 2.12 (Rank-two and rank-k as superpositions). A rank-two matrix is a sum $\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{u}_2\mathbf{v}_2^\top$ that is not itself rank 0 or 1. In general, Theorem 2.7 says rank-k matrices are exactly the sums of k rank-one matrices that cannot be written with fewer. The SVD (Chapter 7) will hand us a *canonical* such decomposition in which the rank-one pieces are mutually orthogonal and ordered by importance.

Machine-learning interpretation

Low rank is the mathematical face of *structure* or *redundancy* in data. A rank-one matrix $\mathbf{u}\mathbf{v}^\top$ is the cleanest possible structure: one "pattern" $\mathbf{u}$ modulated across columns by the weights in $\mathbf{v}$. When a data matrix is approximately a sum of a few rank-one pieces, it is approximately low rank, and the whole apparatus of low-rank approximation (compression, denoising, completion) becomes available.

Summary

Span, independence, basis, and dimension give meaning to the four fundamental subspaces. Rank has three equivalent definitions — column rank, row rank, and minimal factorization dimension — all proved equal in Theorem 2.7. Rank-one matrices are exactly the outer products $\mathbf{u}\mathbf{v}^\top$ (Theorem 2.10), and rank-k matrices are minimal sums of k such products.

Exercise 2.1. Prove that $\operatorname{rank}(AB) \le \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$ using the factorization characterization r_{fac}.

Exercise 2.2. Show that for $A = \mathbf{u}\mathbf{v}^\top$ (nonzero) the null space has dimension $n-1$, and identify it explicitly in terms of $\mathbf{v}$.

Exercise 2.3. Give an example of 2×2 matrices with $\operatorname{rank}(A) = \operatorname{rank}(B) = 1$ but $\operatorname{rank}(A+B) = 2$, and another pair with $\operatorname{rank}(A+B) = 1$. Conclude that rank is neither additive nor subadditive in an exact sense (only the inequality $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$ holds).

Exercise 2.4. Prove the inequality $\operatorname{rank}(A+B) \le \operatorname{rank}(A) + \operatorname{rank}(B)$ from Theorem 2.7.

Chapter 3

Orthogonality

Motivation & intuition

Orthogonality is what makes the SVD *clean*. Without it, decomposing a matrix into ingredients would give pieces that overlap and interfere; with it, the pieces are mutually perpendicular, energies simply add (Pythagoras), and projecting onto a subspace is the optimal way to approximate. Every optimality statement in this book ultimately rests on the geometry of right angles.

3.1 Inner product, norm, orthogonality

Definition 3.1 (Inner product and norm)**.** For $\mathbf{x},\mathbf{y} \in \mathbb{R}^n$ the *inner (dot) product* is $\langle \mathbf{x}, \mathbf{y}\rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$, and the *Euclidean norm* is $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle} = \left(\sum_i x_i^2\right)^{1/2}$. Vectors $\mathbf{x},\mathbf{y}$ are *orthogonal*, written $\mathbf{x} \perp \mathbf{y}$, if $\langle \mathbf{x}, \mathbf{y}\rangle = 0$.

Proposition 3.2 (Basic properties of the inner product)**.** *For all* $\mathbf{x},\mathbf{y},\mathbf{z} \in \mathbb{R}^n$ *and* $c \in \mathbb{R}$*:* (i) *symmetry* $\langle \mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{y}, \mathbf{x}\rangle$*;* (ii) *linearity* $\langle c\mathbf{x}+\mathbf{z}, \mathbf{y}\rangle = c\langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{z}, \mathbf{y}\rangle$*;* (iii) *positivity* $\langle \mathbf{x}, \mathbf{x}\rangle \geq 0$ *with equality iff* $\mathbf{x} = \mathbf{0}$*;* (iv) *the adjoint identity* $\langle A\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, A^\top \mathbf{y}\rangle$ *for any* $A \in \mathbb{R}^{m\times n}$*.*

Proof. (i)–(iii) are immediate from $\langle \mathbf{x}, \mathbf{y}\rangle = \sum_i x_i y_i$. For (iv), $\langle A\mathbf{x}, \mathbf{y}\rangle = (A\mathbf{x})^\top \mathbf{y} = \mathbf{x}^\top A^\top \mathbf{y} = \langle \mathbf{x}, A^\top \mathbf{y}\rangle$, using $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$. ■

Theorem 3.3 (Cauchy–Schwarz inequality)**.** *For all* $\mathbf{x},\mathbf{y} \in \mathbb{R}^n$*,* $|\langle \mathbf{x}, \mathbf{y}\rangle| \leq \|\mathbf{x}\|\,\|\mathbf{y}\|$*, with equality iff* $\mathbf{x},\mathbf{y}$ *are linearly dependent.*

Proof. If $\mathbf{y} = \mathbf{0}$ both sides are 0. Otherwise, for every $t \in \mathbb{R}$ positivity gives

$$0 \leq \|\mathbf{x} - t\mathbf{y}\|^2 = \langle \mathbf{x} - t\mathbf{y}, \mathbf{x} - t\mathbf{y}\rangle = \|\mathbf{x}\|^2 - 2t\langle \mathbf{x}, \mathbf{y}\rangle + t^2\|\mathbf{y}\|^2 =: q(t).$$

The quadratic $q(t)$ is nonnegative for all t, so its discriminant is ≤ 0: $4\langle \mathbf{x}, \mathbf{y}\rangle^2 - 4\|\mathbf{x}\|^2\|\mathbf{y}\|^2 \leq 0$, i.e. $\langle \mathbf{x}, \mathbf{y}\rangle^2 \leq \|\mathbf{x}\|^2\|\mathbf{y}\|^2$. Taking square roots gives the inequality. Equality forces $q(t_0) = 0$ for the vertex t_0, i.e. $\mathbf{x} = t_0\mathbf{y}$, linear dependence; the converse is a direct check. ■

Theorem 3.4 (Pythagorean theorem)**.** *If* $\mathbf{x} \perp \mathbf{y}$ *then* $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$*.*

Proof. $\|\mathbf{x}+\mathbf{y}\|^2 = \langle \mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y}\rangle = \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y}\rangle + \|\mathbf{y}\|^2$, and the middle term vanishes because $\langle \mathbf{x}, \mathbf{y}\rangle = 0$. ■

3.2 Orthonormal sets and orthogonal matrices

Definition 3.5 (Orthonormal set)**.** Vectors $\mathbf{q}_1, \ldots, \mathbf{q}_k$ are *orthonormal* if $\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}$ (unit length, mutually orthogonal).

Proposition 3.6 (Orthonormal $\Rightarrow$ independent)**.** *An orthonormal set is linearly independent.*

Proof. Suppose $\sum_i c_i \mathbf{q}_i = \mathbf{0}$. Take the inner product with $\mathbf{q}_j$: $0 = \langle \mathbf{0}, \mathbf{q}_j \rangle = \sum_i c_i \langle \mathbf{q}_i, \mathbf{q}_j \rangle = \sum_i c_i \delta_{ij} = c_j$. Thus every $c_j = 0$. ∎

Definition 3.7 (Orthogonal matrix)**.** A square matrix $Q \in \mathbb{R}^{n\times n}$ is *orthogonal* if $Q^\top Q = I$. Equivalently its columns are orthonormal.

Proposition 3.8 (Properties of orthogonal matrices)**.** *Let $Q \in \mathbb{R}^{n\times n}$ be orthogonal. Then* (i) *Q is invertible with $Q^{-1} = Q^\top$, hence also $QQ^\top = I$ and the rows are orthonormal;* (ii) *Q preserves inner products: $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$;* (iii) *$Q$ preserves norms: $\|Q\mathbf{x}\| = \|\mathbf{x}\|$;* (iv) *the product of orthogonal matrices is orthogonal.*

Proof. (i) $Q^\top Q = I$ exhibits $Q^\top$ as a left inverse of a square matrix, which is therefore the two-sided inverse, so $QQ^\top = I$ as well; reading $QQ^\top = I$ columnwise says the rows of Q are orthonormal. (ii) $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = (Q\mathbf{x})^\top (Q\mathbf{y}) = \mathbf{x}^\top Q^\top Q\mathbf{y} = \mathbf{x}^\top \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle$. (iii) Put $\mathbf{y} = \mathbf{x}$ in (ii) and take square roots. (iv) If $Q_1^\top Q_1 = I = Q_2^\top Q_2$ then $(Q_1Q_2)^\top (Q_1Q_2) = Q_2^\top Q_1^\top Q_1 Q_2 = Q_2^\top Q_2 = I$. ∎

Geometric interpretation

An orthogonal matrix is a *rigid motion* of space: a rotation, a reflection, or a combination. It moves points around without ever stretching, shrinking, or shearing. Property (iii) is the precise statement that lengths are untouched, and (ii) that angles are untouched. This is why orthogonal factors in the SVD carry no “size” information — all the scaling lives in the diagonal Σ.

3.3 Orthogonal projection onto a subspace

Definition 3.9 (Orthogonal projector)**.** Let $W \subseteq \mathbb{R}^m$ be a subspace with orthonormal basis $\mathbf{q}_1, \ldots, \mathbf{q}_d$, collected as columns of $Q \in \mathbb{R}^{m\times d}$ (so $Q^\top Q = I_d$). The *orthogonal projector* onto W is $P = QQ^\top = \sum_{j=1}^d \mathbf{q}_j \mathbf{q}_j^\top \in \mathbb{R}^{m\times m}$.

Proposition 3.10 (Defining properties of a projector)**.** *With $P = QQ^\top$ as above:* (i) *$P^\top = P$ (symmetric);* (ii) *$P^2 = P$ (idempotent);* (iii) *$P\mathbf{w} = \mathbf{w}$ for $\mathbf{w} \in W$ and $P\mathbf{x} = \mathbf{0}$ for $\mathbf{x} \perp W$;* (iv) *P does not depend on the choice of orthonormal basis of W.*

Proof. (i) $(QQ^\top)^\top = QQ^\top$. (ii) $P^2 = QQ^\top QQ^\top = Q(Q^\top Q)Q^\top = QI_dQ^\top = QQ^\top = P$. (iii) For $\mathbf{w} = Q\mathbf{c} \in W$, $P\mathbf{w} = QQ^\top Q\mathbf{c} = Q\mathbf{c} = \mathbf{w}$; for $\mathbf{x} \perp W$ each $\mathbf{q}_j^\top \mathbf{x} = 0$, so $Q^\top \mathbf{x} = \mathbf{0}$ and $P\mathbf{x} = \mathbf{0}$. (iv) Any $\mathbf{x} \in \mathbb{R}^m$ splits as $\mathbf{x} = \mathbf{w} + \mathbf{x}^\perp$ with $\mathbf{w} \in W$, $\mathbf{x}^\perp \perp W$ (existence of such a split is the orthogonal decomposition, shown next in Theorem 3.11); by (iii) $P\mathbf{x} = \mathbf{w}$, which is determined by W and $\mathbf{x}$ alone, not by the basis. ∎

Theorem 3.11 (Best approximation / orthogonal decomposition)**.** *Let $W \subseteq \mathbb{R}^m$ be a subspace with orthogonal projector P. For every $\mathbf{y} \in \mathbb{R}^m$:*

(a) *$\mathbf{y} - P\mathbf{y} \perp W$, so $\mathbf{y} = P\mathbf{y} + (\mathbf{y} - P\mathbf{y})$ is an orthogonal decomposition;*

(b) *for every $\mathbf{w} \in W$, $\|\mathbf{y} - \mathbf{w}\|^2 = \|\mathbf{y} - P\mathbf{y}\|^2 + \|P\mathbf{y} - \mathbf{w}\|^2$;*

(c) *consequently* $\min_{\mathbf{w}\in W}\|\mathbf{y}-\mathbf{w}\| = \|\mathbf{y}-P\mathbf{y}\|$, *with the unique minimizer* $\mathbf{w} = P\mathbf{y}$.

Proof. (a) For each basis vector $\mathbf{q}_j$ of W, $\langle \mathbf{y}-P\mathbf{y}, \mathbf{q}_j\rangle = \langle \mathbf{y}, \mathbf{q}_j\rangle - \langle QQ^\top\mathbf{y}, \mathbf{q}_j\rangle = \langle \mathbf{y}, \mathbf{q}_j\rangle - (Q^\top\mathbf{y})^\top(Q^\top\mathbf{q}_j)$. Now $Q^\top\mathbf{q}_j = \mathbf{e}_j$, so $(Q^\top\mathbf{y})^\top\mathbf{e}_j = (Q^\top\mathbf{y})_j = \mathbf{q}_j^\top\mathbf{y} = \langle \mathbf{y}, \mathbf{q}_j\rangle$, and the difference is 0. Hence $\mathbf{y}-P\mathbf{y}$ is orthogonal to every basis vector, thus to all of W. (b) Write $\mathbf{y}-\mathbf{w} = (\mathbf{y}-P\mathbf{y}) + (P\mathbf{y}-\mathbf{w})$. The first summand is $\perp W$ by (a), and the second lies in W; they are orthogonal, so Pythagoras (Theorem 3.4) gives the displayed identity. (c) In (b) the first term is fixed and the second is ≥ 0, minimized (uniquely) when $P\mathbf{y}-\mathbf{w} = \mathbf{0}$, i.e. $\mathbf{w} = P\mathbf{y}$. ■

Corollary 3.12 (Matrix Pythagoras)**.** *For any orthogonal projector* $P \in \mathbb{R}^{m\times m}$ *and any* $A \in \mathbb{R}^{m\times n}$,

$$\|A\|_F^2 = \|PA\|_F^2 + \|(I-P)A\|_F^2 .$$

Proof. Apply Theorem 3.11(b) with $\mathbf{w} = P\mathbf{a}_j$ to each column $\mathbf{a}_j$ of A: $\left\|\mathbf{a}_j\right\|^2 = \left\|P\mathbf{a}_j\right\|^2 + \left\|\mathbf{a}_j - P\mathbf{a}_j\right\|^2$. The Frobenius norm squared is the sum of squared column norms (proved in Chapter 8, but here we only use the definition $\|A\|_F^2 = \sum_j \left\|\mathbf{a}_j\right\|^2$), so summing over j gives the claim, noting $(I-P)\mathbf{a}_j = \mathbf{a}_j - P\mathbf{a}_j$. ■

Common misconception

"Projection just means dropping the last coordinates." That is only true when W is a coordinate subspace and the basis is the standard one. In general the projector is $P = QQ^\top$ for an orthonormal basis Q of W; it is the *nearest-point map* onto W, and the direction "dropped" is the orthogonal complement of W, not any fixed set of axes.

Summary

Inner products give length, angle, Cauchy–Schwarz (Theorem 3.3), and Pythagoras (Theorem 3.4). Orthonormal sets are automatically independent (Proposition 3.6); orthogonal matrices preserve all geometry (Proposition 3.8). The orthogonal projector $P = QQ^\top$ realizes the nearest-point map (Theorem 3.11), and the matrix Pythagoras identity (Corollary 3.12) will drive the optimality proof in Chapter 9.

Exercise 3.1. Prove the triangle inequality $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ from Cauchy–Schwarz.

Exercise 3.2. Show that a square matrix preserves all norms (i.e. $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$) if and only if it is orthogonal. *Hint:* polarization recovers the inner product from the norm.

Exercise 3.3. Let P be symmetric and idempotent ($P^\top = P = P^2$). Prove P is the orthogonal projector onto $\mathrm{col}(P)$, and that $I-P$ is the projector onto $\mathrm{col}(P)^\perp$.

Exercise 3.4. Compute the projector onto $W = \mathrm{span}\{(1,1,0)^\top, (0,1,1)^\top\} \subseteq \mathbb{R}^3$ by first orthonormalizing the basis (Gram–Schmidt), then forming $QQ^\top$.

Chapter 4

Symmetric Matrices

Motivation & intuition

The SVD will be manufactured out of the symmetric matrices $A^\top A$ and $AA^\top$. Symmetric matrices are the best-behaved objects in linear algebra: their eigenvalues are real, their eigenvectors can be chosen orthonormal, and they are diagonalized by an *orthogonal* change of basis. This chapter proves the two facts we need and the next chapter assembles them into the Spectral Theorem.

Definition 4.1 (Symmetric matrix)**.** $A \in \mathbb{R}^{n\times n}$ is *symmetric* if $A^\top = A$, i.e. $a_{ij} = a_{ji}$ for all i, j.

Lemma 4.2 (Symmetry is self-adjointness)**.** *A is symmetric iff $\langle A\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, A\mathbf{y}\rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.*

Proof. By Proposition 3.2(iv), $\langle A\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, A^\top\mathbf{y}\rangle$ always. If $A = A^\top$ this equals $\langle \mathbf{x}, A\mathbf{y}\rangle$. Conversely, if $\langle \mathbf{x}, A^\top\mathbf{y}\rangle = \langle \mathbf{x}, A\mathbf{y}\rangle$ for all $\mathbf{x}, \mathbf{y}$ then $\langle \mathbf{x}, (A^\top - A)\mathbf{y}\rangle = 0$ for all $\mathbf{x}$, forcing $(A^\top - A)\mathbf{y} = \mathbf{0}$ for all $\mathbf{y}$, hence $A^\top = A$. ■

Theorem 4.3 (Real eigenvalues)**.** *Every eigenvalue of a real symmetric matrix A is real, and a corresponding eigenvector may be taken real.*

Proof. Allow complex vectors and use the Hermitian inner product $\langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{C}} = \bar{\mathbf{x}}^\top\mathbf{y}$. Suppose $A\mathbf{z} = \lambda\mathbf{z}$ with $\mathbf{z} \neq \mathbf{0}$, $\lambda \in \mathbb{C}$. Then

$$\bar{\mathbf{z}}^\top A\mathbf{z} = \lambda\, \bar{\mathbf{z}}^\top\mathbf{z} = \lambda\, \|\mathbf{z}\|^2 .$$

Taking the conjugate transpose of the scalar $\bar{\mathbf{z}}^\top A\mathbf{z}$ and using A real symmetric ($\bar{A} = A$, $A^\top = A$),

$$\overline{\bar{\mathbf{z}}^\top A\mathbf{z}} = \mathbf{z}^\top A\bar{\mathbf{z}} = (\overline{A\mathbf{z}})^\top\mathbf{z} = \overline{\lambda}\, \bar{\mathbf{z}}^\top\mathbf{z} = \overline{\lambda}\, \|\mathbf{z}\|^2 .$$

A scalar equals its own conjugate, so $\lambda\, \|\mathbf{z}\|^2 = \overline{\lambda}\, \|\mathbf{z}\|^2$; since $\|\mathbf{z}\|^2 > 0$, $\lambda = \overline{\lambda}$, i.e. $\lambda \in \mathbb{R}$. With λ real, $(A - \lambda I)$ is a real singular matrix, so it has a real null vector, providing a real eigenvector. ■

Theorem 4.4 (Orthogonality of eigenvectors for distinct eigenvalues)**.** *If A is symmetric and $A\mathbf{x} = \lambda\mathbf{x}$, $A\mathbf{y} = \mu\mathbf{y}$ with $\lambda \neq \mu$, then $\mathbf{x} \perp \mathbf{y}$.*

Proof. Using Lemma 4.2 two ways,

$$\lambda\langle \mathbf{x}, \mathbf{y}\rangle = \langle \lambda\mathbf{x}, \mathbf{y}\rangle = \langle A\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, A\mathbf{y}\rangle = \langle \mathbf{x}, \mu\mathbf{y}\rangle = \mu\langle \mathbf{x}, \mathbf{y}\rangle.$$

Hence $(\lambda - \mu)\langle \mathbf{x}, \mathbf{y}\rangle = 0$. As $\lambda \neq \mu$, $\langle \mathbf{x}, \mathbf{y}\rangle = 0$. ■

Remark 4.5. Theorem 4.4 handles *distinct* eigenvalues. When an eigenvalue is repeated, its eigenspace is multi-dimensional and we are free to *choose* an orthonormal basis within it (Gram–Schmidt). The Spectral Theorem of the next chapter packages both cases into a single orthonormal eigenbasis of all of $\mathbb{R}^n$.

Summary

Symmetry equals self-adjointness (Lemma 4.2). Real symmetric matrices have real eigenvalues (Theorem 4.3) and orthogonal eigenvectors across distinct eigenvalues (Theorem 4.4). These are exactly the ingredients the Spectral Theorem needs.

Exercise 4.1. Show that if A is symmetric then $\mathbf{x}^{\mathsf{T}}A\mathbf{x}$ is real for all real $\mathbf{x}$ (trivially), and that $\mathbf{x}^{\mathsf{T}}A\mathbf{x} = \sum_i \lambda_i(\mathbf{q}_i^{\mathsf{T}}\mathbf{x})^2$ once the Spectral Theorem is available.

Exercise 4.2. Give a 2×2 *non*-symmetric matrix with complex eigenvalues, illustrating that Theorem 4.3 genuinely uses symmetry.

Exercise 4.3. Prove that for symmetric A, eigenvectors from a repeated eigenvalue need not be orthogonal as given, but can always be *re*chosen orthonormal.

Chapter 5

The Spectral Theorem

Motivation & intuition

The Spectral Theorem is the climax of the symmetric-matrix story and the foundation of the SVD. It says a symmetric matrix is, after an orthogonal change of coordinates, simply a diagonal matrix: it stretches space along n mutually perpendicular axes, each by a real factor, with no rotation mixing the axes. We prove it in full, by induction, using nothing beyond the previous chapter and the fact that a continuous function on a compact set attains its maximum.

5.1 Statement

Theorem 5.1 (Spectral Theorem for real symmetric matrices)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then there exist an orthogonal matrix $Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_n]$ and real numbers $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ such that*

$$A = Q\Lambda Q^{\mathsf{T}} = \sum_{i=1}^{n} \lambda_i \mathbf{q}_i \mathbf{q}_i^{\mathsf{T}}, \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \qquad A\mathbf{q}_i = \lambda_i \mathbf{q}_i.$$

That is, A has an orthonormal basis of eigenvectors with real eigenvalues.

5.2 A lemma: a maximizer of the Rayleigh quotient is an eigenvector

Lemma 5.2 (Existence of a real eigenpair)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric. The function $\mathbf{x} \mapsto \mathbf{x}^{\mathsf{T}} A\mathbf{x}$ attains a maximum on the unit sphere $S = \{\mathbf{x} : \|\mathbf{x}\| = 1\}$ at some $\mathbf{q} \in S$, and that maximizer satisfies $A\mathbf{q} = \lambda \mathbf{q}$ with $\lambda = \mathbf{q}^{\mathsf{T}} A\mathbf{q}$.*

Proof. The map $\mathbf{x} \mapsto \mathbf{x}^{\mathsf{T}} A\mathbf{x}$ is a polynomial, hence continuous, and S is closed and bounded (compact); therefore the maximum is attained at some $\mathbf{q} \in S$, with value $\lambda = \mathbf{q}^{\mathsf{T}} A\mathbf{q}$. We show $\mathbf{q}$ is an eigenvector by a first-order (Lagrange) argument made elementary. Let $\mathbf{v}$ be any vector with $\mathbf{v} \perp \mathbf{q}$, and consider the unit-speed curve $\mathbf{x}(t) = \cos(t)\,\mathbf{q} + \sin(t)\,\hat{\mathbf{v}}$ on S, where $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$ (so $\|\mathbf{x}(t)\| = 1$). Define $g(t) = \mathbf{x}(t)^{\mathsf{T}} A\mathbf{x}(t)$. Since $t = 0$ maximizes g, $g'(0) = 0$. Compute, using symmetry so that $\frac{d}{dt}\mathbf{x}^{\mathsf{T}} A\mathbf{x} = 2\mathbf{x}^{\mathsf{T}} A\dot{\mathbf{x}}$:

$$g'(t) = 2\,\mathbf{x}(t)^{\mathsf{T}} A\,\dot{\mathbf{x}}(t), \qquad \dot{\mathbf{x}}(0) = \hat{\mathbf{v}}, \quad \mathbf{x}(0) = \mathbf{q},$$

so $0 = g'(0) = 2\,\mathbf{q}^{\mathsf{T}} A\hat{\mathbf{v}} = 2\langle A\mathbf{q}, \hat{\mathbf{v}}\rangle$. Thus $A\mathbf{q} \perp \mathbf{v}$ for *every* $\mathbf{v} \perp \mathbf{q}$, which means $A\mathbf{q}$ has no component orthogonal to $\mathbf{q}$: $A\mathbf{q} \in \operatorname{span}\{\mathbf{q}\}$, say $A\mathbf{q} = \lambda\mathbf{q}$. Finally $\lambda = \mathbf{q}^{\mathsf{T}} A\mathbf{q}$ by taking the inner product with $\mathbf{q}$. ∎

5.3 Proof of the Spectral Theorem

Proof of Theorem 5.1. We induct on n. For $n=1$ every 1×1 matrix is diagonal and the claim is trivial with $Q=[1]$.

Assume the theorem for dimension $n-1$. Given symmetric $A\in\mathbb{R}^{n\times n}$, Lemma 5.2 supplies a unit eigenvector $\mathbf{q}_1$ with real eigenvalue $\lambda_1=\mathbf{q}_1^\top A\mathbf{q}_1$, chosen as the *maximal* value of the Rayleigh quotient. Let $W=\mathbf{q}_1^\perp=\{\mathbf{x}:\mathbf{q}_1^\top\mathbf{x}=0\}$, an $(n-1)$-dimensional subspace.

W is A-invariant. If $\mathbf{x}\in W$ then, using symmetry (Lemma 4.2),

$$\langle\mathbf{q}_1,A\mathbf{x}\rangle=\langle A\mathbf{q}_1,\mathbf{x}\rangle=\lambda_1\langle\mathbf{q}_1,\mathbf{x}\rangle=0,$$

so $A\mathbf{x}\in W$. Choose an orthonormal basis $\mathbf{w}_2,\dots,\mathbf{w}_n$ of W and form the orthogonal matrix $R=[\mathbf{q}_1\ \mathbf{w}_2\ \cdots\ \mathbf{w}_n]$. Because W is A-invariant and $\mathbf{q}_1$ is an eigenvector, the matrix of A in this basis is block diagonal:

$$R^\top AR=\begin{bmatrix}\lambda_1 & \mathbf{0}^\top\\ \mathbf{0} & A'\end{bmatrix},$$

where $A'\in\mathbb{R}^{(n-1)\times(n-1)}$ represents $A|_W$. The off-diagonal blocks vanish: the top row is $\mathbf{q}_1^\top A\mathbf{w}_j=\lambda_1\mathbf{q}_1^\top\mathbf{w}_j=0$, and the first column is $\mathbf{w}_j^\top A\mathbf{q}_1=\lambda_1\mathbf{w}_j^\top\mathbf{q}_1=0$. Moreover A' is symmetric, since $R^\top AR$ is symmetric (A is) and a principal submatrix of a symmetric matrix is symmetric.

By the inductive hypothesis $A'=Q'\Lambda'Q'^\top$ with Q' orthogonal and Λ' real diagonal. Set

$$Q=R\begin{bmatrix}1 & \mathbf{0}^\top\\ \mathbf{0} & Q'\end{bmatrix}.$$

Then Q is a product of orthogonal matrices, hence orthogonal (Proposition 3.8(iv)), and

$$Q^\top AQ=\begin{bmatrix}1 & \mathbf{0}^\top\\ \mathbf{0} & Q'^\top\end{bmatrix}(R^\top AR)\begin{bmatrix}1 & \mathbf{0}^\top\\ \mathbf{0} & Q'\end{bmatrix}=\begin{bmatrix}\lambda_1 & \mathbf{0}^\top\\ \mathbf{0} & Q'^\top A'Q'\end{bmatrix}=\begin{bmatrix}\lambda_1 & \mathbf{0}^\top\\ \mathbf{0} & \Lambda'\end{bmatrix}=:\Lambda,$$

a real diagonal matrix. Hence $A=Q\Lambda Q^\top$. Reordering the columns of Q and the diagonal of Λ so that $\lambda_1\ge\cdots\ge\lambda_n$ keeps Q orthogonal and completes the induction. ∎

Corollary 5.3 (Orthogonal diagonalization and the variational characterization)**.** *With the ordering $\lambda_1\ge\cdots\ge\lambda_n$ of Theorem 5.1,*

$$\lambda_1=\max_{\|\mathbf{x}\|=1}\mathbf{x}^\top A\mathbf{x},\qquad \lambda_n=\min_{\|\mathbf{x}\|=1}\mathbf{x}^\top A\mathbf{x},$$

and for any $\mathbf{x}$, writing $\mathbf{x}=\sum_i c_i\mathbf{q}_i$ with $c_i=\mathbf{q}_i^\top\mathbf{x}$,

$$\mathbf{x}^\top A\mathbf{x}=\sum_{i=1}^n\lambda_i c_i^2,\qquad \|\mathbf{x}\|^2=\sum_i c_i^2.$$

Proof. The expansions follow from $A\mathbf{q}_i=\lambda_i\mathbf{q}_i$ and orthonormality. The extremal statements follow because $\sum_i\lambda_i c_i^2$ is a weighted average of the λ_i with weights $c_i^2\ge 0$ summing to $\|\mathbf{x}\|^2=1$, maximized by putting all weight on the largest λ_1 (achieved at $\mathbf{x}=\mathbf{q}_1$) and minimized at $\mathbf{q}_n$. ∎

Geometric interpretation

Theorem 5.1 says: rotate space by $Q^\top$ so that the eigenvectors become the coordinate axes; in those coordinates A is the diagonal matrix Λ, pure axiswise scaling; then rotate back by Q. A symmetric

> matrix is a stretch along n perpendicular axes — nothing more.

Summary

A maximizer of the Rayleigh quotient on the unit sphere is an eigenvector (Lemma 5.2); deflating onto its orthogonal complement, which is A-invariant, and inducting proves the Spectral Theorem (Theorem 5.1): every real symmetric matrix is orthogonally diagonalizable with real eigenvalues. Corollary 5.3 records the variational characterization that we will generalize (Ky Fan) in Chapter 9.

Exercise 5.1. Prove the *min–max* (Courant–Fischer) formula for λ_k as a corollary of the Spectral Theorem.

Exercise 5.2. Show $\operatorname{tr}(A) = \sum_i \lambda_i$ and $\det(A) = \prod_i \lambda_i$ for symmetric A.

Exercise 5.3. Diagonalize $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ orthogonally and verify $A = Q\Lambda Q^\top$ explicitly.

Chapter 6

Positive Semidefinite Matrices

Motivation & intuition

The SVD is built from $A^\top A$ and $AA^\top$. These are not arbitrary symmetric matrices — they are *positive semidefinite*, which forces their eigenvalues to be *nonnegative*. That nonnegativity is exactly what lets us take square roots and define the singular values $\sigma_i = \sqrt{\lambda_i}$. This short chapter proves the three facts the SVD construction will quote verbatim.

Definition 6.1 (Positive semidefinite)**.** A symmetric matrix $M \in \mathbb{R}^{n\times n}$ is *positive semidefinite* (PSD), written $M \succeq 0$, if $\mathbf{x}^\top M\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$. It is *positive definite* ($M \succ 0$) if $\mathbf{x}^\top M\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Proposition 6.2 ($A^\top A$ and $AA^\top$ are symmetric)**.** *For any $A \in \mathbb{R}^{m\times n}$, both $A^\top A \in \mathbb{R}^{n\times n}$ and $AA^\top \in \mathbb{R}^{m\times m}$ are symmetric.*

Proof. $(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$, using $(XY)^\top = Y^\top X^\top$ and $(A^\top)^\top = A$. Likewise $(AA^\top)^\top = (A^\top)^\top A^\top = AA^\top$. ■

Proposition 6.3 ($A^\top A$ and $AA^\top$ are PSD)**.** *For any $A \in \mathbb{R}^{m\times n}$, $A^\top A \succeq 0$ and $AA^\top \succeq 0$. Hence (by the Spectral Theorem) all their eigenvalues are ≥ 0.*

Proof. For any $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^\top (A^\top A)\mathbf{x} = (A\mathbf{x})^\top (A\mathbf{x}) = \|A\mathbf{x}\|^2 \geq 0$, so $A^\top A \succeq 0$. Replacing A by $A^\top$ gives $AA^\top \succeq 0$. If $M\mathbf{q} = \lambda\mathbf{q}$ with $\|\mathbf{q}\| = 1$ and $M \succeq 0$, then $\lambda = \mathbf{q}^\top M\mathbf{q} \geq 0$. ■

Proposition 6.4 (Shared nonzero spectrum of $A^\top A$ and $AA^\top$)**.** *The matrices $A^\top A$ and $AA^\top$ have the same nonzero eigenvalues, with the same multiplicities. Specifically, if $A^\top A\,\mathbf{v} = \lambda\mathbf{v}$ with $\lambda \neq 0$, then $\mathbf{u} := A\mathbf{v}$ satisfies $AA^\top\mathbf{u} = \lambda\mathbf{u}$ and $\mathbf{u} \neq \mathbf{0}$.*

Proof. From $A^\top A\mathbf{v} = \lambda\mathbf{v}$, left-multiply by A: $AA^\top(A\mathbf{v}) = A(A^\top A\mathbf{v}) = A(\lambda\mathbf{v}) = \lambda(A\mathbf{v})$, i.e. $AA^\top\mathbf{u} = \lambda\mathbf{u}$. Moreover $\|A\mathbf{v}\|^2 = \mathbf{v}^\top A^\top A\mathbf{v} = \lambda\,\|\mathbf{v}\|^2 \neq 0$, so $\mathbf{u} = A\mathbf{v} \neq \mathbf{0}$. The map $\mathbf{v} \mapsto A\mathbf{v}$ thus carries each λ-eigenvector of $A^\top A$ to a λ-eigenvector of $AA^\top$; it is injective on that eigenspace (if $A\mathbf{v} = \mathbf{0}$ then $\lambda\mathbf{v} = A^\top A\mathbf{v} = \mathbf{0}$ forces $\mathbf{v} = \mathbf{0}$), so multiplicities match. By symmetry of roles, the correspondence is a bijection on nonzero eigenspaces. ■

Machine-learning interpretation

Proposition 6.3 is the reason PCA works at all: the covariance matrix $\Sigma = \frac{1}{m}X_c^\top X_c$ is of the form $A^\top A$, hence symmetric PSD, hence has real nonnegative eigenvalues $\lambda_i \geq 0$ that we interpret as *variances*. A negative variance would be nonsense; PSD-ness guarantees it never happens. Proposition 6.4 is the engine of the "Gram-matrix trick": when there are far fewer samples than features, one eigendecomposes the small $m \times m$ matrix $AA^\top$ and transfers the eigenvectors to the large space via $\mathbf{v} \mapsto A\mathbf{v}$.

Summary

$A^\top A$ and $AA^\top$ are always symmetric (Proposition 6.2) and PSD (Proposition 6.3), so their eigenvalues are real and nonnegative — permitting the square roots $\sigma_i = \sqrt{\lambda_i}$ that define singular values. The two products share their nonzero spectrum (Proposition 6.4), the link that will tie left and right singular vectors together.

Exercise 6.1. Prove $A^\top A \succ 0$ (positive *definite*) iff A has linearly independent columns (equivalently $\mathrm{null}(A) = \{\mathbf{0}\}$).

Exercise 6.2. Show that $M \succeq 0$ iff $M = B^\top B$ for some matrix B. *Hint:* use the Spectral Theorem to build $B = \Lambda^{1/2} Q^\top$.

Exercise 6.3. For $A = \left(\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right)$, compute $A^\top A$ and $AA^\top$ and verify they share the eigenvalues $\{45, 5\}$.

Chapter 7

Derivation of the Singular Value Decomposition

Motivation & intuition

We now *build* the SVD; we do not assume it. The plan: $A^\top A$ is symmetric PSD (Chapter 6), so the Spectral Theorem (Chapter 5) gives it an orthonormal eigenbasis with nonnegative eigenvalues. We declare those eigenvectors to be the right singular vectors, the square roots of the eigenvalues to be the singular values, and we *derive* the left singular vectors as the normalized images $A\mathbf{v}_i/\sigma_i$. Verifying orthonormality of the left vectors and reassembling proves $A = U\Sigma V^\top$. Because the construction only ever uses the Spectral Theorem applied to a matrix that always exists, the SVD always exists.

7.1 Construction

Theorem 7.1 (Existence of the SVD). *Every $A \in \mathbb{R}^{m\times n}$ admits a factorization*

$$A = U\Sigma V^\top,$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal and $\Sigma \in \mathbb{R}^{m\times n}$ has entries $\Sigma_{ii} = \sigma_i \geq 0$ for $i = 1,\dots,\min(m,n)$ and zeros off the diagonal, with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots$, where $r = \operatorname{rank}(A)$.

Proof. **Right singular vectors and singular values.** By Propositions 6.2–6.3, $A^\top A \in \mathbb{R}^{n\times n}$ is symmetric PSD. The Spectral Theorem (Theorem 5.1) gives an orthonormal basis $\mathbf{v}_1,\dots,\mathbf{v}_n$ of $\mathbb{R}^n$ with

$$A^\top A\,\mathbf{v}_i = \lambda_i \mathbf{v}_i, \qquad \lambda_1 \geq \cdots \geq \lambda_n \geq 0.$$

Define $\sigma_i = \sqrt{\lambda_i} \geq 0$. Let r be the number of strictly positive σ_i (we show $r = \operatorname{rank} A$ below). Set $V = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$, orthogonal.

Left singular vectors. For $i = 1,\dots,r$ (so $\sigma_i > 0$) define

$$\mathbf{u}_i := \frac{1}{\sigma_i} A\mathbf{v}_i \in \mathbb{R}^m.$$

These are orthonormal: for $i, j \leq r$,

$$\mathbf{u}_i^\top \mathbf{u}_j = \frac{1}{\sigma_i\sigma_j}(A\mathbf{v}_i)^\top (A\mathbf{v}_j) = \frac{1}{\sigma_i\sigma_j}\mathbf{v}_i^\top (A^\top A\mathbf{v}_j) = \frac{\lambda_j}{\sigma_i\sigma_j}\mathbf{v}_i^\top \mathbf{v}_j = \frac{\sigma_j^2}{\sigma_i\sigma_j}\delta_{ij} = \frac{\sigma_j}{\sigma_i}\delta_{ij} = \delta_{ij}.$$

Extend $\mathbf{u}_1,\dots,\mathbf{u}_r$ to an orthonormal basis $\mathbf{u}_1,\dots,\mathbf{u}_m$ of $\mathbb{R}^m$ (Gram–Schmidt on any completion), and set $U=[\mathbf{u}_1\ \cdots\ \mathbf{u}_m]$, orthogonal.

The annihilated directions. For $i>r$ we have $\lambda_i=0$, so $\|A\mathbf{v}_i\|^2=\mathbf{v}_i^\top A^\top A\mathbf{v}_i=\lambda_i=0$, hence $A\mathbf{v}_i=\mathbf{0}$.

Reassembly. Define $\Sigma\in\mathbb{R}^{m\times n}$ by $\Sigma_{ii}=\sigma_i$ and zeros elsewhere. We verify $A=U\Sigma V^\top$ by checking both sides agree on the basis $\mathbf{v}_1,\dots,\mathbf{v}_n$ (two linear maps equal on a basis are equal, Lemma 2.4). The right side sends

$$(U\Sigma V^\top)\mathbf{v}_j=U\Sigma\mathbf{e}_j=U(\sigma_j\mathbf{e}_j)=\sigma_j\mathbf{u}_j\quad(j\le r),\qquad (U\Sigma V^\top)\mathbf{v}_j=\sigma_j\mathbf{u}_j=\mathbf{0}\quad(j>r,\ \sigma_j=0),$$

using $V^\top\mathbf{v}_j=\mathbf{e}_j$. The left side sends $A\mathbf{v}_j=\sigma_j\mathbf{u}_j$ for $j\le r$ (by definition of $\mathbf{u}_j$) and $A\mathbf{v}_j=\mathbf{0}$ for $j>r$. The two maps agree on the basis, so $A=U\Sigma V^\top$.

$r=\operatorname{rank}A$**.** From $A=U\Sigma V^\top$ with U,V invertible, $\operatorname{rank}(A)=\operatorname{rank}(\Sigma)$ (multiplying by invertible matrices preserves rank, Exercise), and Σ has exactly r nonzero entries on its diagonal, so $\operatorname{rank}(\Sigma)=r$. ∎

Corollary 7.2 (Outer-product / dyadic form)**.** *With the notation of Theorem 7.1,*

$$A=\sum_{i=1}^{r}\sigma_i\,\mathbf{u}_i\mathbf{v}_i^\top.$$

Each $\mathbf{u}_i\mathbf{v}_i^\top$ is rank one (Theorem 2.10) and the terms are ordered by $\sigma_1\ge\cdots\ge\sigma_r>0$.

Proof. Multiplying out $U\Sigma V^\top$ keeps only the r nonzero diagonal entries of Σ, each pairing column $\mathbf{u}_i$ of U with row $\mathbf{v}_i^\top$ of $V^\top$, weighted by σ_i. ∎

Corollary 7.3 (U and V as eigenvectors; the EVD link)**.** *The columns of V are orthonormal eigenvectors of $A^\top A$ with eigenvalues σ_i^2, and the columns of U are orthonormal eigenvectors of $AA^\top$ with the same nonzero eigenvalues:*

$$A^\top A=V\Sigma^\top\Sigma V^\top,\qquad AA^\top=U\Sigma\Sigma^\top U^\top.$$

Proof. $A^\top A=(U\Sigma V^\top)^\top(U\Sigma V^\top)=V\Sigma^\top U^\top U\Sigma V^\top=V(\Sigma^\top\Sigma)V^\top$, using $U^\top U=I$. This is the orthogonal diagonalization of $A^\top A$, so the columns of V are its eigenvectors with eigenvalues $(\Sigma^\top\Sigma)_{ii}=\sigma_i^2$. Symmetrically $AA^\top=U(\Sigma\Sigma^\top)U^\top$. ∎

7.2 Full, reduced, and compact SVD

Definition 7.4 (Three forms)**.** Let $r=\operatorname{rank}A$.

- **Full SVD:** $A=U\Sigma V^\top$ with $U\in\mathbb{R}^{m\times m}$, $V\in\mathbb{R}^{n\times n}$ orthogonal and $\Sigma\in\mathbb{R}^{m\times n}$.
- **Reduced (thin) SVD:** keep the first $\min(m,n)$ columns of U and the corresponding block of Σ.
- **Compact SVD:** keep only the r columns with $\sigma_i>0$: $A=U_r\Sigma_rV_r^\top$, $U_r\in\mathbb{R}^{m\times r}$, $\Sigma_r\in\mathbb{R}^{r\times r}$ (invertible diagonal), $V_r\in\mathbb{R}^{n\times r}$, $U_r^\top U_r=I_r=V_r^\top V_r$.

Proposition 7.5 (The four fundamental subspaces from the SVD)**.** *With the compact SVD $A=U_r\Sigma_rV_r^\top$,*

$$\operatorname{col}(A)=\operatorname{span}\{\mathbf{u}_1,\dots,\mathbf{u}_r\},\qquad \operatorname{null}(A^\top)=\operatorname{span}\{\mathbf{u}_{r+1},\dots,\mathbf{u}_m\},$$

$$\operatorname{row}(A)=\operatorname{span}\{\mathbf{v}_1,\dots,\mathbf{v}_r\},\qquad \operatorname{null}(A)=\operatorname{span}\{\mathbf{v}_{r+1},\dots,\mathbf{v}_n\}.$$

Proof. $A\mathbf{x}=\sum_{i\le r}\sigma_i(\mathbf{v}_i^\top\mathbf{x})\mathbf{u}_i$ lies in $\operatorname{span}\{\mathbf{u}_i\}_{i\le r}$, and every $\mathbf{u}_i$ ($i\le r$) is attained (take $\mathbf{x}=\mathbf{v}_i$), so $\operatorname{col}(A)=\operatorname{span}\{\mathbf{u}_i\}_{i\le r}$. For the null space, $A\mathbf{x}=\mathbf{0}$ iff $\sigma_i(\mathbf{v}_i^\top\mathbf{x})=0$ for all $i\le r$ iff $\mathbf{v}_i^\top\mathbf{x}=0$ for $i\le r$ (as $\sigma_i>0$), i.e. $\mathbf{x}\in\operatorname{span}\{\mathbf{v}_i\}_{i>r}$. The $A^\top$ statements follow by applying these to $A^\top=V\Sigma^\top U^\top$. ∎

7.3 Uniqueness

Theorem 7.6 (Uniqueness of the SVD). *The singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$ are uniquely determined by A (they are the square roots of the nonzero eigenvalues of $A^\top A$). If the nonzero singular values are* distinct, *then each pair $(\mathbf{u}_i, \mathbf{v}_i)$ is unique up to a common sign $(\mathbf{u}_i, \mathbf{v}_i) \mapsto (-\mathbf{u}_i, -\mathbf{v}_i)$. If a singular value is repeated, the corresponding singular vectors are determined only up to an orthonormal rotation within the shared eigenspace.*

Proof. The σ_i^2 are the eigenvalues of the matrix $A^\top A$, which depends only on A; eigenvalues are unique, so the σ_i are. For a simple (multiplicity-one) eigenvalue σ_i^2 of $A^\top A$, the eigenspace is one-dimensional, so $\mathbf{v}_i$ is determined up to sign; then $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ inherits the same sign choice, giving the common sign freedom. For a repeated eigenvalue, the Spectral Theorem only fixes the eigenspace, within which any orthonormal basis is admissible; the relation $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ transports that freedom to the $\mathbf{u}_i$. ■

Example 7.7 (A complete 2×2 SVD by hand). Let $A = \left(\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right)$. Then $A^\top A = \left(\begin{smallmatrix} 25 & 20 \\ 20 & 25 \end{smallmatrix}\right)$, with characteristic equation $(25-\lambda)^2 - 400 = 0$, so $\lambda = 25 \pm 20$, giving $\lambda_1 = 45$, $\lambda_2 = 5$ and $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$. The eigenvectors of $A^\top A$ are $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$ (for 45) and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$ (for 5). Then

$$\mathbf{u}_1 = \frac{A\mathbf{v}_1}{\sigma_1} = \frac{1}{3\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 3 \\ 9 \end{smallmatrix}\right) = \frac{1}{\sqrt{10}} \left(\begin{smallmatrix} 1 \\ 3 \end{smallmatrix}\right), \qquad \mathbf{u}_2 = \frac{A\mathbf{v}_2}{\sigma_2} = \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 3 \\ -1 \end{smallmatrix}\right) = \frac{1}{\sqrt{10}} \left(\begin{smallmatrix} 3 \\ -1 \end{smallmatrix}\right).$$

These $\mathbf{u}_i$ are exactly the eigenvectors of $AA^\top = \left(\begin{smallmatrix} 9 & 12 \\ 12 & 41 \end{smallmatrix}\right)$ (eigenvalues $45, 5$), confirming Corollary 7.3. One checks $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^\top$ reproduces A.

Geometric interpretation

Reading $A = U\Sigma V^\top$ right to left: $V^\top$ *rotates* the input so the right singular vectors become coordinate axes; Σ *scales* axis i by σ_i (and embeds/projects between $\mathbb{R}^n$ and $\mathbb{R}^m$); U *rotates* the result into the output space along the left singular vectors. Consequently A maps the unit sphere of $\mathbb{R}^n$ onto an ellipsoid in $\mathbb{R}^m$ whose semi-axes have lengths σ_i in directions $\mathbf{u}_i$.

Common misconception

"The SVD is just the eigendecomposition of A." No: A need not be square or diagonalizable, and even when it is, the singular values are *not* the absolute values of the eigenvalues in general (they coincide only for symmetric PSD A). The SVD is the eigendecomposition of the symmetric PSD matrices $A^\top A$ and $AA^\top$, stitched together — a strictly more general and always-available object.

Summary

Applying the Spectral Theorem to $A^\top A$ yields V and the σ_i; defining $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ yields an orthonormal U; reassembly gives $A = U\Sigma V^\top$ (Theorem 7.1), equivalently the dyadic sum (Corollary 7.2). V, U diagonalize $A^\top A, AA^\top$ (Corollary 7.3); the SVD reads off all four fundamental subspaces (Proposition 7.5); it is unique up to the sign / rotation freedom of Theorem 7.6.

Exercise 7.1. Prove that multiplying by invertible (in particular orthogonal) matrices preserves rank, the fact quoted at the end of Theorem 7.1.

Exercise 7.2. Show the pseudoinverse $A^+ := V_r \Sigma_r^{-1} U_r^\top$ satisfies $AA^+A = A$ and $A^+AA^+ = A^+$.

Exercise 7.3. For A symmetric PSD, prove the SVD and the EVD coincide ($U = V = Q$, $\sigma_i = \lambda_i$).

Exercise 7.4. Compute the full SVD of $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right)$.

Chapter 8

Matrix Norms

Motivation & intuition

To say one matrix is the *best* approximation of another we must measure the size of their difference. Two norms matter. The *Frobenius* norm treats a matrix as a long vector and measures total entrywise energy; it is the natural least-squares error. The *spectral* norm measures the largest stretch A applies to any unit vector; it is the natural worst-case error. The Eckart–Young–Mirsky theorem holds for *both*, and we prove the identities we need for each.

8.1 The Frobenius norm and trace

Definition 8.1 (Trace)**.** For square M, $\operatorname{tr}(M) = \sum_i M_{ii}$.

Lemma 8.2 (Cyclic invariance of trace)**.** *For conformable matrices,* $\operatorname{tr}(XY) = \operatorname{tr}(YX)$*, and hence* $\operatorname{tr}(XYZ) = \operatorname{tr}(YZX) = \operatorname{tr}(ZXY)$.

Proof. $\operatorname{tr}(XY) = \sum_i \sum_j X_{ij} Y_{ji} = \sum_j \sum_i Y_{ji} X_{ij} = \operatorname{tr}(YX)$. The three-factor identity follows by grouping $X(YZ)$ etc. ■

Definition 8.3 (Frobenius norm)**.** $\|A\|_F = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2 \right)^{1/2}$ for $A \in \mathbb{R}^{m \times n}$.

Proposition 8.4 (Frobenius via trace)**.** $\|A\|_F^2 = \operatorname{tr}(A^\top A) = \operatorname{tr}(AA^\top) = \sum_j \left\|\mathbf{a}_j\right\|^2$*, where* $\mathbf{a}_j$ *are the columns.*

Proof. $(A^\top A)_{jj} = \sum_i a_{ij}^2 = \left\|\mathbf{a}_j\right\|^2$; summing over j gives $\operatorname{tr}(A^\top A) = \sum_{i,j} a_{ij}^2 = \|A\|_F^2$. The equality $\operatorname{tr}(A^\top A) = \operatorname{tr}(AA^\top)$ is Lemma 8.2. ■

Theorem 8.5 (Orthogonal invariance of the Frobenius norm)**.** *If* $Q \in \mathbb{R}^{m \times m}$*,* $P \in \mathbb{R}^{n \times n}$ *are orthogonal, then for all* $A \in \mathbb{R}^{m \times n}$*,* $\|QAP\|_F = \|A\|_F$.

Proof. Using Proposition 8.4 and the cyclic property,

$$\|QAP\|_F^2 = \operatorname{tr}\big((QAP)^\top QAP\big) = \operatorname{tr}\big(P^\top A^\top Q^\top QAP\big) = \operatorname{tr}\big(A^\top A\, PP^\top\big) = \operatorname{tr}(A^\top A) = \|A\|_F^2\,,$$

where $Q^\top Q = I$ and $PP^\top = I$. ■

Corollary 8.6 (Frobenius norm in singular values)**.** $\|A\|_F^2 = \sum_{i=1}^{r} \sigma_i^2$.

Proof. By Theorem 8.5 with the SVD $A = U\Sigma V^\top$, $\|A\|_F = \left\|U\Sigma V^\top\right\|_F = \|\Sigma\|_F$, and $\|\Sigma\|_F^2 = \sum_i \sigma_i^2$. ■

8.2 The spectral (operator-2) norm

Definition 8.7 (Spectral norm)**.** $\|A\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|A\mathbf{x}\|}{\|\mathbf{x}\|} = \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$.

Theorem 8.8 (Spectral norm equals the top singular value)**.** *$\|A\|_2 = \sigma_1$, the largest singular value, with the maximum attained at $\mathbf{x} = \mathbf{v}_1$ (and $A\mathbf{v}_1 = \sigma_1 \mathbf{u}_1$).*

Proof. For $\|\mathbf{x}\| = 1$ write $\mathbf{x} = \sum_i c_i \mathbf{v}_i$ with $\sum_i c_i^2 = 1$ (Corollary 5.3 applied to the orthonormal basis V). Then $A\mathbf{x} = \sum_{i \leq r} \sigma_i c_i \mathbf{u}_i$, and since the $\mathbf{u}_i$ are orthonormal,

$$\|A\mathbf{x}\|^2 = \sum_{i \leq r} \sigma_i^2 c_i^2 \leq \sigma_1^2 \sum_i c_i^2 = \sigma_1^2,$$

so $\|A\mathbf{x}\| \leq \sigma_1$. Equality holds at $\mathbf{x} = \mathbf{v}_1$ (then $c_1 = 1$, rest 0), where $\|A\mathbf{v}_1\| = \sigma_1$. Hence the maximum is exactly σ_1. ■

Remark 8.9. Both norms are written entirely in singular values: $\|A\|_F^2 = \sum_i \sigma_i^2$ and $\|A\|_2 = \sigma_1$. This is why the SVD is the right tool for approximation questions phrased in either norm.

Summary

The Frobenius norm is total entrywise energy, equal to $\operatorname{tr}(A^\top A)$ (Proposition 8.4), invariant under orthogonal multiplication (Theorem 8.5), and equal to $\sqrt{\sum_i \sigma_i^2}$ (Corollary 8.6). The spectral norm is the maximal stretch, equal to σ_1 (Theorem 8.8). Both are functions of the singular values alone.

Exercise 8.1. Prove $\|A\|_2 \leq \|A\|_F \leq \sqrt{r}\,\|A\|_2$ where $r = \operatorname{rank} A$.

Exercise 8.2. Show $\|AB\|_F \leq \|A\|_2 \|B\|_F$ and $\|AB\|_F \leq \|A\|_F \|B\|_2$.

Exercise 8.3. Prove $\|A\|_2 = \|A^\top\|_2$ and $\|A\|_F = \|A^\top\|_F$ directly from the SVD.

Chapter 9

The Best Rank-k Approximation: Eckart–Young–Mirsky

Motivation & intuition

Everything has been building to this. We truncate the dyadic SVD to its first k terms and prove that the result is the closest rank-k matrix to A — not merely a good choice, but the provably optimal one — in both the Frobenius and the spectral norm. The Frobenius proof goes through two lemmas: "best matrix for a fixed column space is a projection" (which converts the problem from choosing a matrix to choosing a subspace), and the *Ky Fan* maximum principle (which solves the subspace problem). The spectral proof uses a short dimension-counting argument. No step is taken on faith.

9.1 The truncated SVD and its error

Definition 9.1 (Truncated SVD)**.** For $A = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ and $1 \le k \le r$, define

$$A_k = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top = U_k \Sigma_k V_k^\top,$$

where U_k, V_k keep the first k columns and $\Sigma_k = \operatorname{diag}(\sigma_1, \dots, \sigma_k)$. By construction $\operatorname{rank}(A_k) \le k$.

Proposition 9.2 (Error of the truncation)**.** $\|A - A_k\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2$ *and* $\|A - A_k\|_2 = \sigma_{k+1}$ *(with* $\sigma_{r+1} := 0$*).*

Proof. $A - A_k = \sum_{i=k+1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ is itself an SVD (orthonormal $\mathbf{u}_i, \mathbf{v}_i$) with singular values $\sigma_{k+1} \ge \cdots \ge \sigma_r$. Apply Corollary 8.6 for the Frobenius statement and Theorem 8.8 for the spectral statement. ■

This computes the error of *our* candidate. The theorem asserts no rank-k matrix does better. We prove that next.

9.2 Lemma A: reduction to a subspace problem

Lemma 9.3 (Best matrix for a fixed column space)**.** *Fix a subspace $S \subseteq \mathbb{R}^m$ with $\dim S \le k$, and let P_S be the orthogonal projector onto S. Among all $B \in \mathbb{R}^{m \times n}$ with $\operatorname{col}(B) \subseteq S$, the Frobenius error $\|A - B\|_F$ is minimized by $B = P_S A$, and*

$$\min_{\operatorname{col}(B) \subseteq S} \|A - B\|_F^2 = \|(I - P_S)A\|_F^2 = \|A\|_F^2 - \|P_S A\|_F^2 .$$

Proof. Every column $\mathbf{b}_j$ of B lies in S. By the best-approximation theorem (Theorem 3.11(c)) applied to the column $\mathbf{a}_j$ of A, $\left\|\mathbf{a}_j - \mathbf{b}_j\right\| \geq \left\|\mathbf{a}_j - P_S\mathbf{a}_j\right\|$ with equality at $\mathbf{b}_j = P_S\mathbf{a}_j$. Squaring and summing over j (Proposition 8.4) gives $\|A-B\|_F^2 \geq \|(I-P_S)A\|_F^2$, attained at $B = P_SA$. The final equality is the matrix Pythagoras identity (Corollary 3.12). ∎

Remark 9.4 (Why this is the crucial reduction). A matrix has rank $\leq k$ *iff* its columns lie in some subspace of dimension $\leq k$ (Theorem 2.7). Lemma 9.3 says that once the subspace S is fixed, the optimal B is forced. Hence

$$\min_{\operatorname{rank}(B)\leq k} \|A-B\|_F^2 = \min_{\dim S\leq k} \left(\|A\|_F^2 - \|P_SA\|_F^2\right) = \|A\|_F^2 - \max_{\dim S\leq k} \|P_SA\|_F^2\,.$$

Minimizing the error is the *same* as maximizing the captured energy $\|P_SA\|_F^2$ over k-dimensional subspaces. That maximization is Lemma B.

9.3 Lemma B: the Ky Fan maximum principle

We first rewrite the captured energy as a sum of quadratic forms.

Lemma 9.5 (Captured energy is a sum of quadratic forms)**.** *If S has orthonormal basis $\mathbf{w}_1,\ldots,\mathbf{w}_d$ (so $P_S = \sum_j \mathbf{w}_j\mathbf{w}_j^\top$), then*

$$\|P_SA\|_F^2 = \sum_{j=1}^{d} \mathbf{w}_j^\top (AA^\top)\mathbf{w}_j.$$

Proof. $\|P_SA\|_F^2 = \operatorname{tr}\big((P_SA)^\top P_SA\big) = \operatorname{tr}(A^\top P_S^\top P_SA) = \operatorname{tr}(A^\top P_SA)$, using $P_S^\top = P_S$ and $P_S^2 = P_S$ (Proposition 3.10). With $P_S = \sum_j \mathbf{w}_j\mathbf{w}_j^\top$ and cyclicity of trace, $\operatorname{tr}(A^\top P_SA) = \sum_j \operatorname{tr}(A^\top\mathbf{w}_j\mathbf{w}_j^\top A) = \sum_j (\mathbf{w}_j^\top A)(A^\top\mathbf{w}_j) = \sum_j \mathbf{w}_j^\top (AA^\top)\mathbf{w}_j$. ∎

Lemma 9.6 (Ky Fan: maximal sum of quadratic forms)**.** *Let $M \in \mathbb{R}^{m\times m}$ be symmetric with eigenvalues $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_m$ and orthonormal eigenvectors $\mathbf{q}_1,\ldots,\mathbf{q}_m$ (Spectral Theorem). For every orthonormal set $\mathbf{w}_1,\ldots,\mathbf{w}_k$,*

$$\sum_{j=1}^{k} \mathbf{w}_j^\top M\mathbf{w}_j \;\leq\; \mu_1 + \cdots + \mu_k,$$

with equality when $\mathbf{w}_j = \mathbf{q}_j$.

Proof. Extend $\mathbf{w}_1,\ldots,\mathbf{w}_k$ to an orthonormal basis $\mathbf{w}_1,\ldots,\mathbf{w}_m$ of $\mathbb{R}^m$. Set $c_{ij} = \mathbf{q}_i^\top\mathbf{w}_j$. Using $M = \sum_i \mu_i\mathbf{q}_i\mathbf{q}_i^\top$,

$$\sum_{j=1}^{k} \mathbf{w}_j^\top M\mathbf{w}_j = \sum_{j=1}^{k}\sum_{i=1}^{m} \mu_i(\mathbf{q}_i^\top\mathbf{w}_j)^2 = \sum_{i=1}^{m} \mu_i t_i, \qquad t_i := \sum_{j=1}^{k} c_{ij}^2.$$

We claim $0 \leq t_i \leq 1$ and $\sum_i t_i = k$:

- $t_i \geq 0$ trivially; and $t_i \leq \sum_{j=1}^m c_{ij}^2 = \|\mathbf{q}_i\|^2 = 1$, because $\{\mathbf{w}_j\}_{j=1}^m$ is a full orthonormal basis so the squared coordinates of the unit vector $\mathbf{q}_i$ sum to 1; truncating to $k \leq m$ terms can only decrease the sum.
- $\sum_{i=1}^m t_i = \sum_{j=1}^k \sum_{i=1}^m c_{ij}^2 = \sum_{j=1}^k \left\|\mathbf{w}_j\right\|^2 = k$, since $\{\mathbf{q}_i\}$ is an orthonormal basis.

We must therefore bound $\sum_i \mu_i t_i$ over $0 \leq t_i \leq 1$, $\sum_i t_i = k$. For any such t,

$$\sum_{i=1}^{m} \mu_i t_i - \sum_{i=1}^{k} \mu_i = \sum_{i=1}^{k} \mu_i(t_i - 1) + \sum_{i>k} \mu_i t_i \leq \mu_k \sum_{i=1}^{k} (t_i - 1) + \mu_k \sum_{i>k} t_i = \mu_k\Big(\sum_i t_i - k\Big) = 0,$$

where the inequality uses $\mu_i \geq \mu_k$ together with $t_i - 1 \leq 0$ on the first sum, and $\mu_i \leq \mu_k$ with $t_i \geq 0$ on the second. Hence $\sum_i \mu_i t_i \leq \sum_{i=1}^{k} \mu_i$. Choosing $\mathbf{w}_j = \mathbf{q}_j$ gives $t_i = 1$ for $i \leq k$ and 0 otherwise, achieving equality. ∎

Geometric interpretation

Ky Fan in words: each unit direction $\mathbf{w}_j$ has a total "budget" of 1 to distribute across the eigen-axes of M; k orthonormal directions distribute a total budget of k, but no single axis can absorb more than 1. To maximize $\sum_i \mu_i t_i$ you place a full unit on each of the k largest eigenvalues — i.e. align your directions with the top k eigenvectors.

9.4 The theorem (Frobenius norm)

Theorem 9.7 (Eckart–Young–Mirsky, Frobenius norm)**.** *Let $A \in \mathbb{R}^{m\times n}$ have singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$ and let $1 \leq k \leq r$. Then for every B with* $\operatorname{rank}(B) \leq k$*,*

$$\|A - B\|_F \;\geq\; \|A - A_k\|_F\,, \qquad \textit{i.e.} \qquad \min_{\operatorname{rank}(B)\leq k} \|A - B\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2,$$

attained at $B = A_k$. If $\sigma_k > \sigma_{k+1}$ the minimizer A_k is unique.

Proof. Let B have $\operatorname{rank}(B) \leq k$ and set $S = \operatorname{col}(B)$, so $\dim S \leq k$. By Lemma 9.3,

$$\|A - B\|_F^2 \;\geq\; \|A\|_F^2 - \|P_S A\|_F^2\,.$$

Enlarging S to dimension exactly k only increases $\|P_S A\|_F^2$ (Lemma 9.5 adds nonnegative terms), so assume $\dim S = k$ with orthonormal basis $\mathbf{w}_1, \ldots, \mathbf{w}_k$. By Lemma 9.5 and the Ky Fan principle (Lemma 9.6) applied to $M = AA^\top$ — whose eigenvalues are $\sigma_1^2 \geq \cdots \geq \sigma_r^2 \geq 0$ with eigenvectors the left singular vectors $\mathbf{u}_i$ (Corollary 7.3) —

$$\|P_S A\|_F^2 = \sum_{j=1}^{k} \mathbf{w}_j^\top (AA^\top)\mathbf{w}_j \;\leq\; \sigma_1^2 + \cdots + \sigma_k^2.$$

Therefore, using $\|A\|_F^2 = \sum_{i=1}^{r} \sigma_i^2$ (Corollary 8.6),

$$\|A - B\|_F^2 \;\geq\; \sum_{i=1}^{r} \sigma_i^2 - \sum_{i=1}^{k} \sigma_i^2 = \sum_{i=k+1}^{r} \sigma_i^2 = \|A - A_k\|_F^2\,,$$

the last equality by Proposition 9.2. Thus no rank-k matrix beats A_k. The bound is attained by A_k: taking $S^\star = \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ makes Ky Fan an equality, and the optimal B from Lemma 9.3 is

$$P_{S^\star} A = U_k U_k^\top A = U_k U_k^\top (U\Sigma V^\top) = U_k \Sigma_k V_k^\top = A_k,$$

because $U_k^\top U$ extracts the first k rows of $\Sigma V^\top$.

Uniqueness when $\sigma_k > \sigma_{k+1}$. Equality throughout forces (i) Ky Fan tight and (ii) the projection bound tight. With the strict gap $\sigma_k^2 > \sigma_{k+1}^2$, the optimal weight vector t in Lemma 9.6 is the unique vertex $t_1 = \cdots = t_k = 1$, which forces $S = \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ exactly; then (ii) forces $B = P_S A = A_k$. ∎

Common misconception

A popular shortcut argues: rotate by U,V to reduce the problem to $\min_{\mathrm{rank}(C)\le k}\|\Sigma - C\|_F$ with $C = U^\top BV$; observe that off-diagonal entries of C only add error; conclude "the optimal C is diagonal," then keep the top k σ_i. The *answer* is right but the conclusion "optimal C is diagonal" is *not* justified by the off-diagonal observation alone: zeroing off-diagonals can change the rank, so one is not comparing matrices of the same rank. Optimizing the diagonal and the off-diagonal separately ignores that the rank constraint couples all entries. The genuine content — that no rank-k competitor (diagonal or not) beats the truncation — is supplied precisely by Ky Fan (Lemma 9.6). Our proof never assumes diagonality; the diagonal form of A_k *emerges* as the conclusion.

9.5 The theorem (spectral norm)

Theorem 9.8 (Eckart–Young–Mirsky, spectral norm)**.** *With the notation above, for every B with* $\mathrm{rank}(B)\le k$*,*

$$\|A-B\|_2 \;\ge\; \sigma_{k+1} = \|A-A_k\|_2 .$$

Hence A_k is also a best rank-k approximation in the spectral norm.

Proof. We already have $\|A-A_k\|_2 = \sigma_{k+1}$ (Proposition 9.2). Let B have $\mathrm{rank}(B)\le k$, so $\dim\mathrm{null}(B)\ge n-k$ (Rank–Nullity). Consider the $(k+1)$-dimensional subspace $T = \mathrm{span}\{\mathbf{v}_1,\dots,\mathbf{v}_{k+1}\}$. Since

$$\dim\mathrm{null}(B)+\dim T \;\ge\; (n-k)+(k+1) = n+1 > n,$$

the two subspaces of $\mathbb{R}^n$ intersect nontrivially; pick a unit vector $\mathbf{w}\in\mathrm{null}(B)\cap T$. Write $\mathbf{w} = \sum_{i=1}^{k+1} c_i\mathbf{v}_i$ with $\sum_i c_i^2 = 1$. Then $B\mathbf{w} = \mathbf{0}$, so $(A-B)\mathbf{w} = A\mathbf{w} = \sum_{i=1}^{k+1}\sigma_i c_i\mathbf{u}_i$, and by orthonormality of the $\mathbf{u}_i$,

$$\|(A-B)\mathbf{w}\|^2 = \sum_{i=1}^{k+1}\sigma_i^2 c_i^2 \;\ge\; \sigma_{k+1}^2\sum_{i=1}^{k+1} c_i^2 = \sigma_{k+1}^2,$$

using $\sigma_i\ge\sigma_{k+1}$ for $i\le k+1$. Therefore $\|A-B\|_2 \ge \|(A-B)\mathbf{w}\|\,/\,\|\mathbf{w}\| \ge \sigma_{k+1}$. ∎

Remark 9.9 (Weyl's inequality, optional). The spectral result is a special case of *Weyl's inequality* $\sigma_{i+j-1}(X+Y)\le\sigma_i(X)+\sigma_j(Y)$: with $X = A-B$, $Y = B$, $j = k+1$, and $\sigma_{k+1}(B) = 0$ (since $\mathrm{rank}\,B\le k$), one gets $\sigma_{i+k}(A)\le\sigma_i(A-B)$, whose $i=1$ case is the theorem and whose squares summed give an alternative proof of the Frobenius version.

Machine-learning interpretation

The two norms answer two questions. Frobenius ($\sum_{i>k}\sigma_i^2$) is the *total* reconstruction energy lost — the right metric for compression and PCA, where average error matters. Spectral (σ_{k+1}) is the *worst-case* direction-wise error — the right metric for stability and for guarantees of the form "no single mode is mis-estimated by more than σ_{k+1}." Remarkably, the same matrix A_k is optimal for both.

Summary

The truncation A_k has Frobenius error $\sqrt{\sum_{i>k}\sigma_i^2}$ and spectral error σ_{k+1} (Proposition 9.2). Lemma 9.3 converts "best rank-k matrix" into "best k-dimensional subspace"; Lemma 9.6 (Ky Fan) solves that subspace problem; together they prove the Frobenius optimality of A_k (Theorem 9.7). A dimension-counting argument proves spectral optimality (Theorem 9.8). In both norms, A_k is best, and the discarded singular values quantify the error exactly.

Exercise 9.1. Prove that if $\sigma_k = \sigma_{k+1}$ the minimizer in Theorem 9.7 need not be unique, by exhibiting a matrix and two distinct optimal A_k's.

Exercise 9.2. Deduce from Theorem 9.7 that the relative error $\|A - A_k\|_F \,/\, \|A\|_F = \sqrt{\sum_{i>k} \sigma_i^2 / \sum_i \sigma_i^2}$ equals one minus the captured "energy fraction."

Exercise 9.3. Using the intersection argument of Theorem 9.8, prove the more general statement $\|A - B\|_2 \geq \sigma_{k+1}$ holds even when B ranges over all matrices with $\operatorname{rank}(B) \leq k$ that need not share A's singular vectors.

Exercise 9.4. Prove the Frobenius theorem a second way from Weyl's inequality (state and assume Weyl).

Chapter 10

Intuition After the Proof: Where the SVD Lives

Motivation & intuition

Now that optimality is *proved*, we may indulge in intuition without risk of circularity. Each application below is the Eckart–Young–Mirsky theorem wearing different clothes: replace a matrix by its best low-rank approximation and read off the consequences.

10.1 Geometry: hyper-ellipsoids and dominant directions

A sends the unit sphere of $\mathbb{R}^n$ to an ellipsoid in $\mathbb{R}^m$ with semi-axis lengths σ_i along $\mathbf{u}_i$ (Chapter 7). Truncating to A_k keeps the k longest axes and flattens the rest — the best k-dimensional "shadow" of the ellipsoid, by Theorem 9.7.

10.2 Principal Component Analysis

Let $X \in \mathbb{R}^{m\times n}$ hold m data points in rows; centre it to $X_c = X - \mathbf{1}\boldsymbol{\mu}^\top$. The covariance $\Sigma_{\text{cov}} = \frac{1}{m}X_c^\top X_c$ is symmetric PSD, and by Corollary 7.3 its eigenvectors are the right singular vectors $\mathbf{v}_i$ of X_c with eigenvalues $\lambda_i = \sigma_i^2/m$. Thus:

$$\textbf{principal directions} = \{\mathbf{v}_i\}, \qquad \textbf{variance along } \mathbf{v}_i = \sigma_i^2/m.$$

Projecting onto $\operatorname{span}\{\mathbf{v}_1,\dots,\mathbf{v}_k\}$ gives the k-dimensional representation of least squared reconstruction error — exactly Theorem 9.7 applied to X_c. PCA *is* truncated SVD of the centred data.

10.3 Image compression

A grayscale image is a matrix of intensities. Its singular values typically decay quickly, so A_k with modest k is visually indistinguishable from A while storing $k(m+n+1)$ numbers instead of mn. The error $\sqrt{\sum_{i>k}\sigma_i^2}$ is the provably smallest achievable at rank k.

10.4 Latent Semantic Analysis and word embeddings

For a term–document matrix A (entry = frequency of term i in document j), the truncated SVD $A_k = U_k \Sigma_k V_k^\top$ maps terms and documents into a shared k-dimensional "concept" space. Synonymous terms, though never co-occurring, acquire nearby rows in $U_k \Sigma_k$ because they load on the same singular directions. Modern word-embedding methods are descendants of this idea: a low-rank factorization of a (transformed) co-occurrence matrix.

10.5 Recommendation systems and matrix completion

A user–item matrix is modelled as approximately low rank: tastes are governed by a few latent factors. Filling missing entries and truncating by SVD predicts unseen ratings; the low-rank assumption is exactly the statement that $A \approx A_k$, whose optimality Theorem 9.7 guarantees among rank-k models.

10.6 LoRA: low-rank adaptation of large models

Fine-tuning a pretrained weight matrix W_0 produces an update ΔW. Empirically ΔW is close to low rank, so LoRA stores it factored as $\Delta W = BA$ with $B \in \mathbb{R}^{m \times k}$, $A \in \mathbb{R}^{k \times n}$, $k \ll m, n$. Training only B, A replaces mn parameters with $k(m+n)$ — the same compression accounting as Chapter 9, deployed on gradient updates instead of images. The viability of LoRA is the empirical claim that the best rank-k approximation of the update loses little, i.e. that the tail $\sum_{i>k} \sigma_i^2(\Delta W)$ is small.

10.7 Why the SVD is central to machine learning

Every task above reduces to the same operation: *represent a matrix by its best low-rank approximation*. The SVD provides that approximation, the theorem certifies its optimality, and the singular values quantify exactly what is kept and what is lost. That combination — a universal decomposition plus a sharp optimality guarantee plus an interpretable error — is why the SVD recurs throughout statistics, signal processing, and deep learning.

Summary

PCA, compression, LSA, embeddings, recommenders, and LoRA are one theorem in disguise: substitute A_k for A. The right singular vectors are PCA's principal directions and LSA's concepts; the singular values are variances, signal strengths, and compression budgets; the discarded tail $\sum_{i>k} \sigma_i^2$ is the provably minimal information lost.

Exercise 10.1. Show that the rank-k PCA reconstruction $\hat{X} = \mathbf{1}\boldsymbol{\mu}^\top + (X_c V_k) V_k^\top$ equals $\mathbf{1}\boldsymbol{\mu}^\top + (X_c)_k$, the truncated SVD of the centred data, and conclude its optimality from Theorem 9.7.

Exercise 10.2. For an image with singular values $\sigma_i \propto i^{-1}$, estimate the relative Frobenius error of A_k as a function of k.

Exercise 10.3. Argue that if a fine-tuning update has rapidly decaying singular values, a small LoRA rank k suffices, and quantify "suffices" via $\sum_{i>k} \sigma_i^2$.

Notation and Theorem Dependency

Notation. $\mathbb{R}^n$ real n-vectors; $A^\top$ transpose; $\langle \cdot, \cdot \rangle$ inner product; $\|\cdot\|$ Euclidean norm; $\|\cdot\|_F$ Frobenius norm; $\|\cdot\|_2$ spectral norm; tr trace; col, row, null the fundamental subspaces; rank rank; P_S orthogonal projector onto S; σ_i singular values; $\mathbf{u}_i, \mathbf{v}_i$ left/right singular vectors; A_k rank-k truncated SVD.

Dependency spine. Theorem 2.7 (rank) and Theorem 2.10 (rank-one) $\rightarrow$ Theorem 3.11 (best approximation) and Corollary 3.12 (matrix Pythagoras) $\rightarrow$ Theorem 4.4 (orthogonal eigenvectors) $\rightarrow$ Theorem 5.1 (Spectral Theorem) $\rightarrow$ Propositions 6.3, 6.4 (PSD) $\rightarrow$ Theorem 7.1 (SVD exists) $\rightarrow$ Theorem 8.5, Corollary 8.6, Theorem 8.8 (norms) $\rightarrow$ Lemmas 9.3, 9.5, 9.6 $\rightarrow$ Theorems 9.7, 9.8 (Eckart–Young–Mirsky).

"The SVD is the Spectral Theorem for rectangular matrices; Eckart–Young–Mirsky says that keeping the largest singular values loses provably the least."